

What-if: setup cost reduced by 50% (SME)

Delta Pumps & Valves Co. (fictional case)

Generated 28 September 2026 | Prepared by Industrial Engineering Intern | Lot-Sizing Tool v1.0.0

Items analysed

22

Annual savings vs current

\$44,516

Avg inventory value (recommended)

\$236,304

Avg inventory value (current)

\$309,125

Orders per year (recommended)

150

Orders per year (current)

180

Savings, current cost and current inventory refer only to items for which a current order quantity was supplied.

Models: EOQ (Wilson), EPQ (finite production rate), all-units quantity discounts. Deterministic, level demand.

How to read this report

Delta Pumps & Valves Co. (fictional case) | 22 items | currency \$

What was calculated

For every item the tool finds the order (or production) quantity Q that minimises the total annual cost = ordering/setup cost ($A \cdot D/Q$) + carrying cost ($Q/2 \cdot v \cdot r$, multiplied by $(1-D/m)$ for produced items) + purchase cost ($D \cdot v$, which only matters when quantity discounts exist).

Models used

EOQ = $\sqrt{2AD / vr}$ for purchased items (entire order arrives at once).

EPQ = $\sqrt{2AD / (vr(1-D/m))}$ for items produced in-house at rate m .

All-units quantity discounts: the optimum of each price tier is found; if it falls below the tier's break quantity the break quantity is used; all tiers are then compared on total cost (lecture Cases 1, 2 and 3).

Practical rules

The theoretical optimum is rounded to the pack size and kept within the minimum and maximum order quantities. The column 'Practical rules applied' shows which rules shaped each answer.

Reorder point

Place an order when the inventory position (on hand + on order) falls to $D \times L$, where L is the lead time. Demand is assumed level and known; safety stock for uncertain demand is NOT included.

Flexibility band

Any order size between 'Flex band low' and 'Flex band high' costs at most 2% more ordering+carrying cost than the optimum. The EOQ cost curve is flat near its minimum ($PCP = 50 a^2/(1+a)$), so rounding to pallets or truckloads inside this band is essentially free.

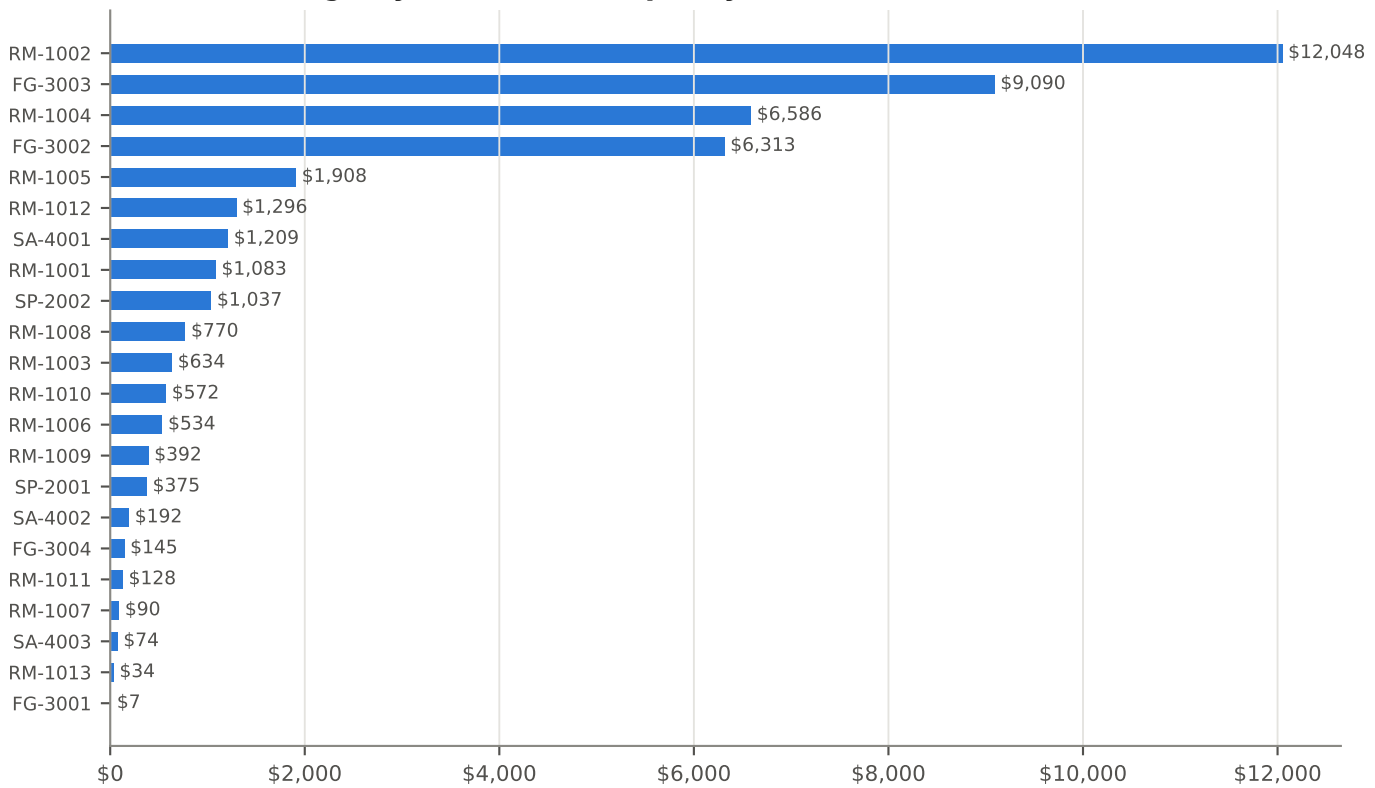
PCP of the current policy

Percentage Cost Penalty = $(\text{cost of current } Q - \text{cost of recommended } Q) / \text{optimal TRC} \times 100$.

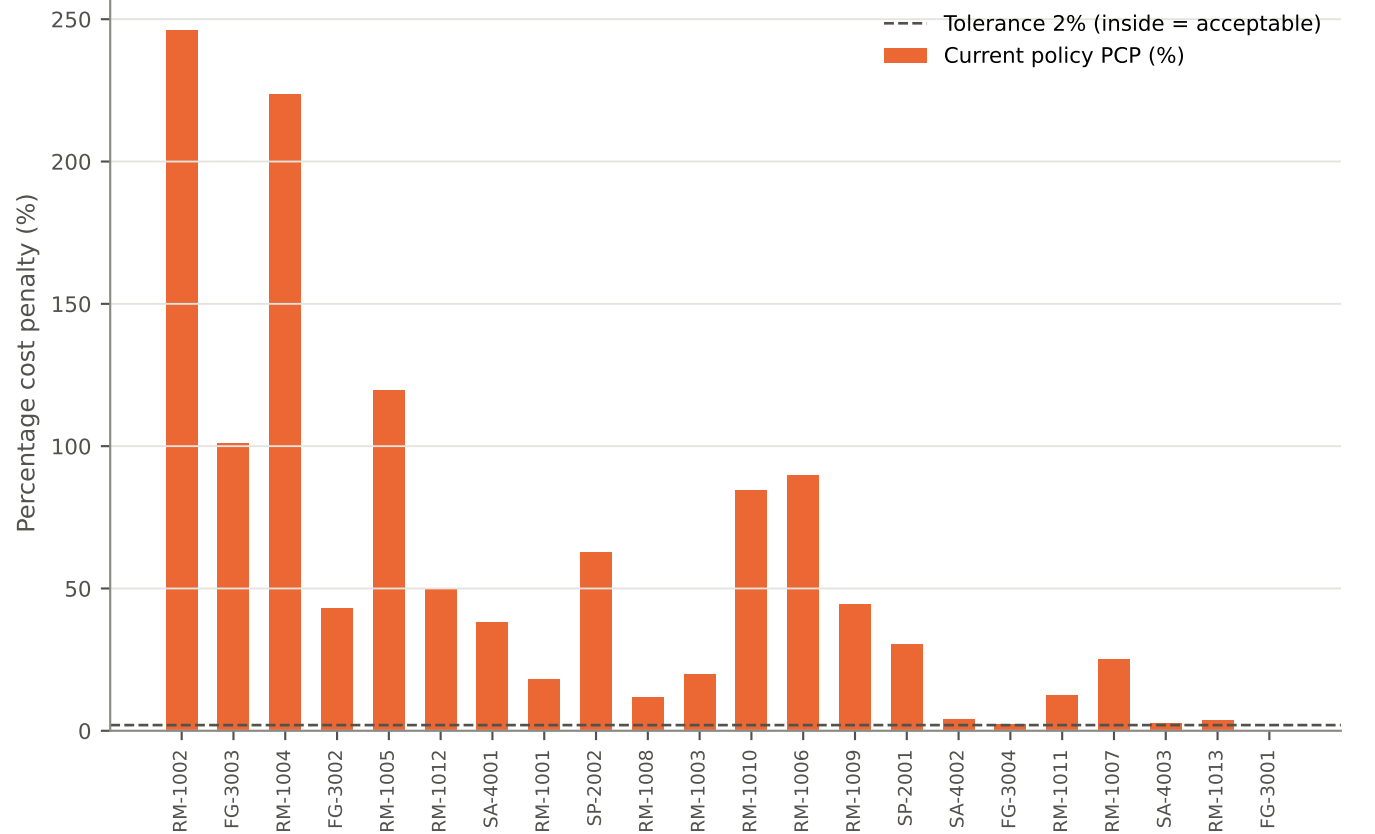
Limitations

Level, deterministic demand; costs constant over time; items treated independently (no joint replenishment); no shortages. Re-run the analysis when inputs change significantly.

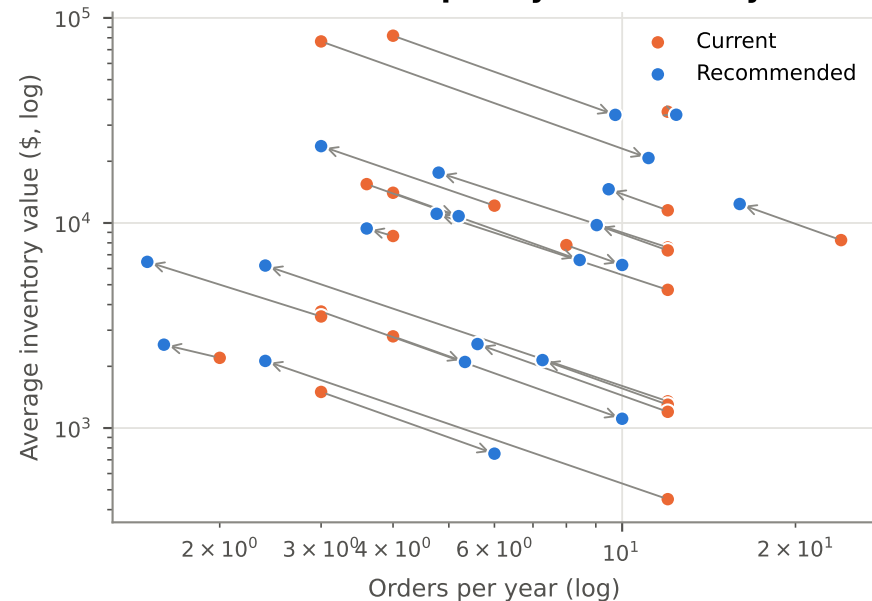
Annual savings by SKU (current policy -> recommended), total \$44,516



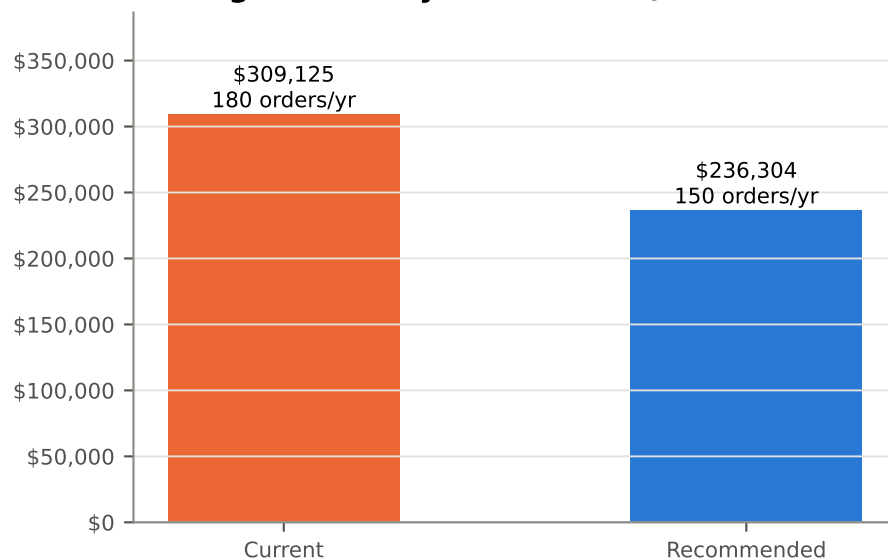
How far is each current policy from optimal? (PCP, SKUs ordered by savings)



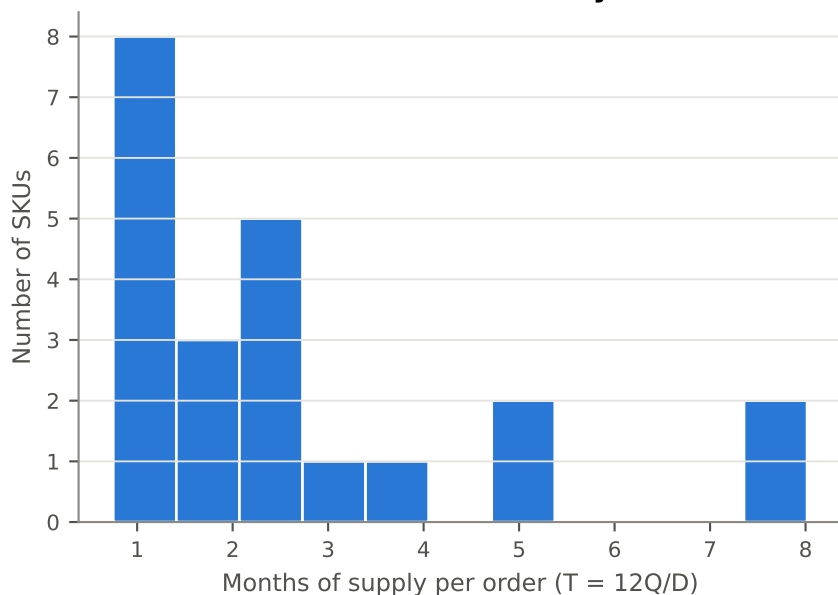
Trade-off: order frequency vs inventory



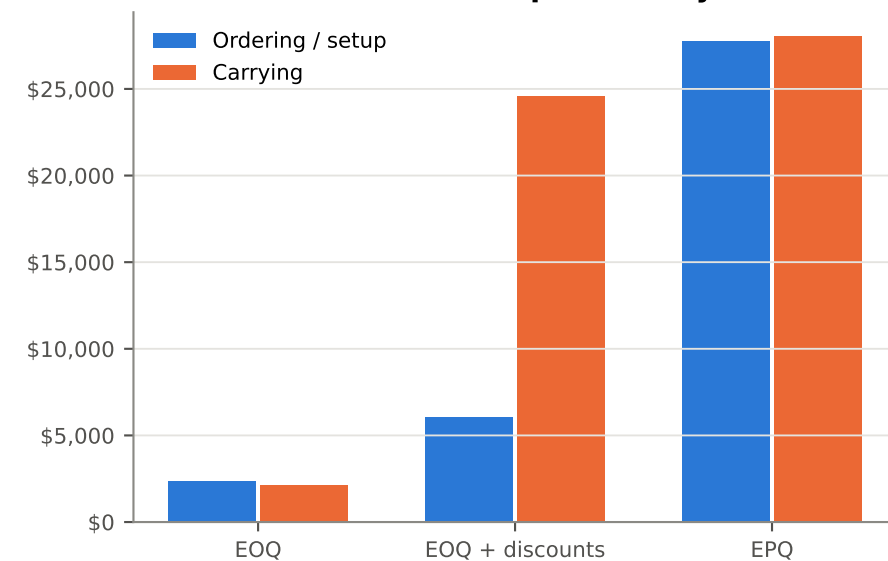
Average inventory investment (items with current policy)



Distribution of recommended cycle times



Recommended TRC components by model



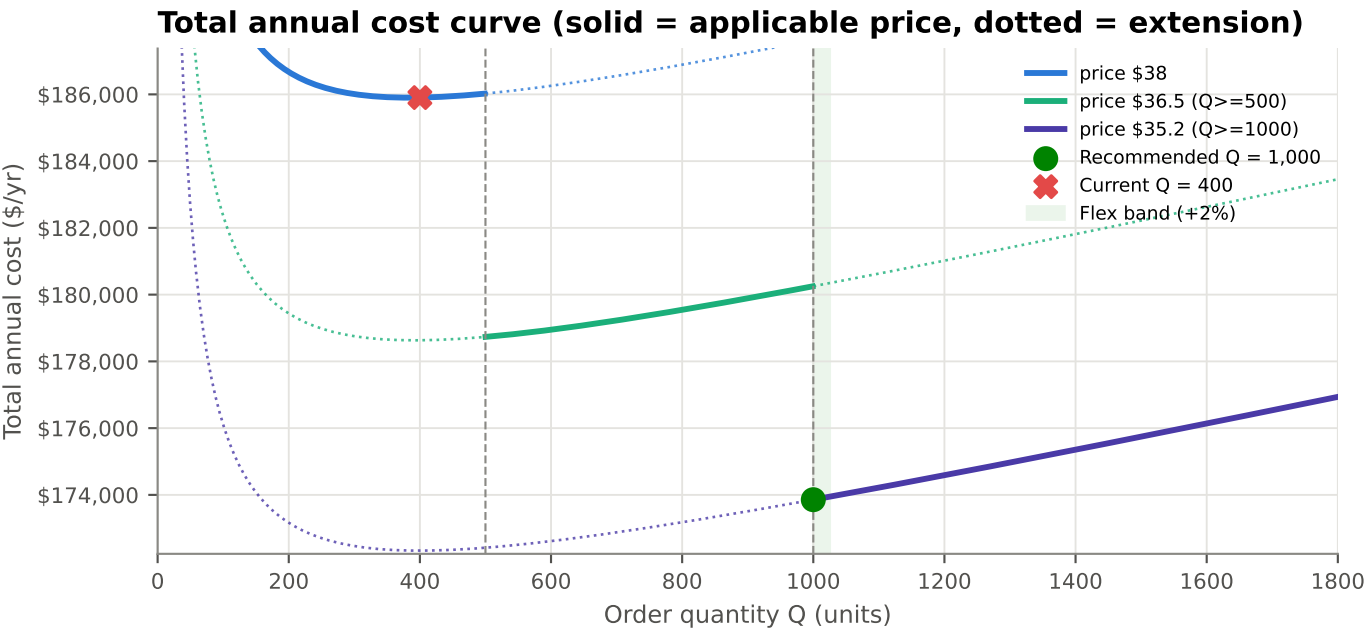
Recommendations - summary table

SKU	Model	Rec. Q	Textbook Q	Price (\$)	Orders/yr	Cycle (days)	ROP	Avg inv (\$)	TRC (\$)	Current Q	Savings (\$)	PCP %
RM-1001	EOQ + discount	12,000	2,464.0	3.950	3.0	83.3	1,440	23,700	5,943	6,000	1,083	18.2
RM-1002	EOQ + discount	1,000	383.9	35.200	4.8	52.1	384	17,600	4,896	400	12,048	246.1
RM-1003	EOQ + discount	1,000	480.7	21.600	5.2	48.1	312	10,800	3,216	1,300	634	19.7
RM-1004	EOQ + discount	2,016	637.8	11.000	4.8	52.5	461	11,088	2,947	800	6,586	223.5
RM-1005	EOQ + discount	10,000	2,582.0	1.240	2.4	104.2	672	6,200	1,596	2,000	1,908	119.5
RM-1006	EOQ + discount	25,000	9,860.1	0.170	2.4	104.2	1,200	2,125	594	5,000	534	89.9
RM-1007	EOQ	25,000	25,000.0	0.060	6.0	41.7	3,000	750	360	50,000	90	25.0
RM-1008	EOQ + discount	150	163.3	165.000	16.0	15.6	288	12,375	6,490	100	770	11.9
RM-1009	EOQ	300	385.5	7.400	10.0	25.0	96	1,110	883	1,000	392	44.4
RM-1010	EOQ + discount	5,000	1,557.0	1.020	1.6	156.2	192	2,550	676	4,000	572	84.6
RM-1011	EOQ	165	164.1	26.000	7.3	34.4	67	2,145	1,024	100	128	12.5
RM-1012	EOQ + discount	2,000	770.6	9.400	3.6	69.4	518	9,400	2,598	1,800	1,296	49.9
SP-2001	EOQ	107	107.0	48.000	5.6	44.6	60	2,568	1,233	50	375	30.4
SP-2002	EOQ + discount	1,002	240.9	12.900	1.5	167.0	90	6,463	1,648	500	1,037	62.9
FG-3001	EPQ	290	290.6	310.000	12.4	20.1	72	33,712	14,865	300	7	0.1
FG-3002	EPQ	185	183.6	455.000	9.7	25.7	36	33,670	14,705	450	6,313	42.9
FG-3003	EPQ	540	533.0	96.000	11.1	22.5	96	20,736	9,006	2,000	9,090	100.9
FG-3004	EPQ	950	929.9	41.000	9.5	26.4	144	14,606	6,292	750	145	2.3
SA-4001	EPQ	640	643.9	27.500	8.4	29.6	65	6,600	3,187	1,500	1,209	37.9
SA-4002	EPQ	465	465.5	63.000	9.0	27.7	50	9,765	4,692	350	192	4.1
SA-4003	EPQ	480	480.4	52.000	10.0	25.0	38	6,240	2,998	600	74	2.5
RM-1013	EOQ	375	382.2	11.200	5.3	46.9	160	2,100	942	500	34	3.6

Input validation log

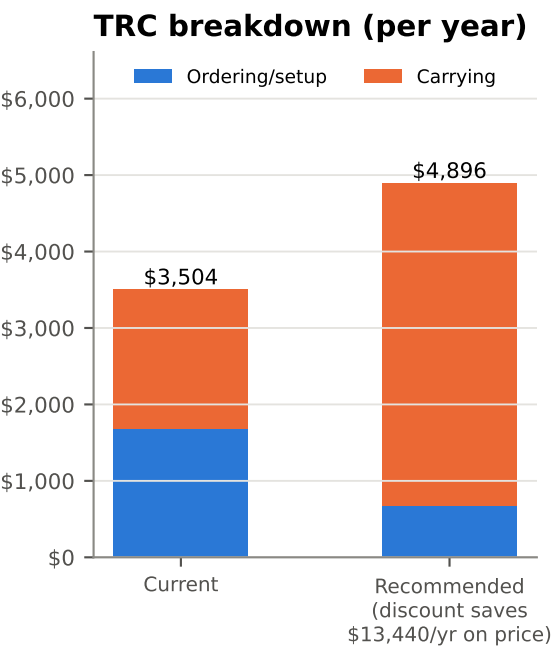
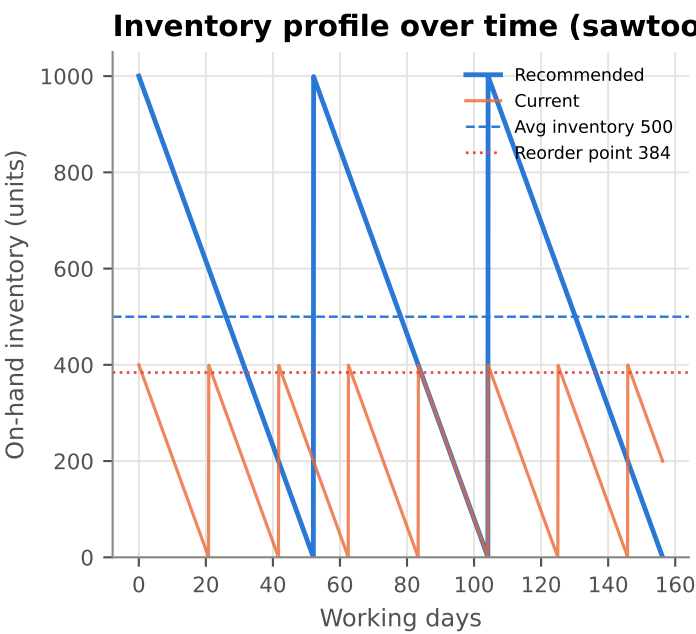
sku	level	message
RM-1013	WARNING	carrying_rate 22.0 > 1 - did you enter a percentage? Interpreted as 0.22
FG-3005	ERROR	production_rate (2000) must exceed annual_demand (2500) - item skipped

RM-1002 - Cast iron pump housing [EOQ + discounts]



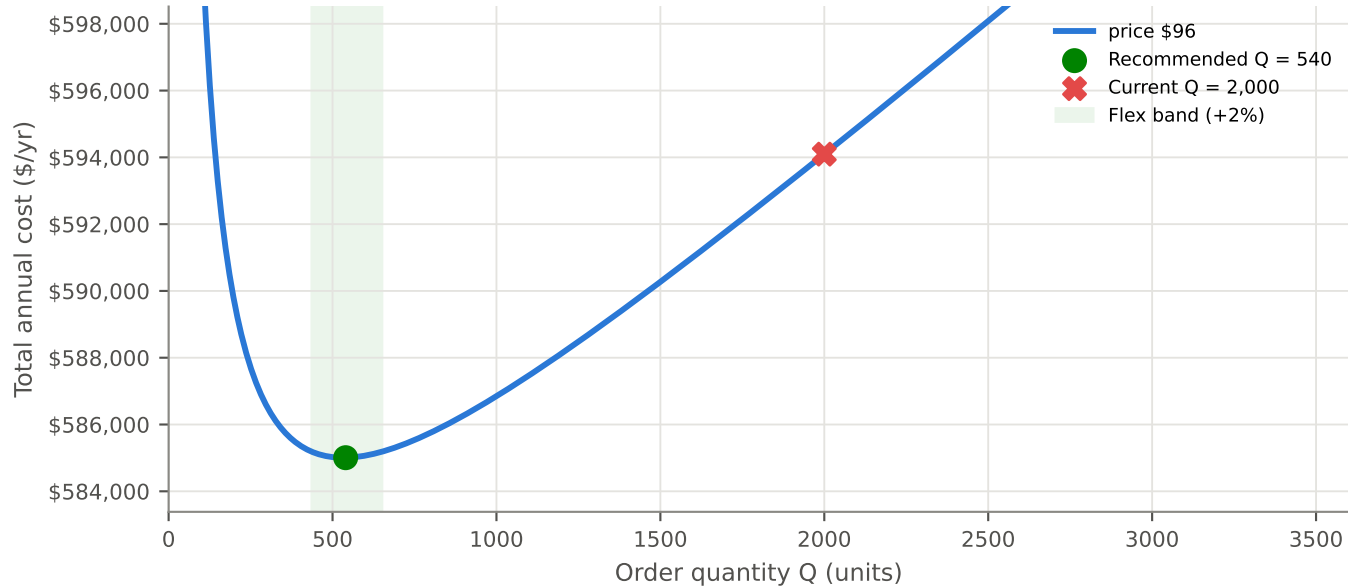
Key figures

Metric	Value
Annual demand D	4,800
Order/setup cost A	\$140.00
Base unit cost v0	\$38.000
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	383.9
Recommended Q	1,000
Price paid	\$35.200
Orders per year	4.8
Cycle (working days)	52.1
Reorder point	384
TRC (order+carry)	\$4,896
Total annual cost	\$173,856
Case / tier	Tier 3 of 3
Rules applied	Pack 10
Current Q	400
Annual savings	\$12,048
Current PCP	246.1%



FG-3003 - Gate valve DN100 [EPQ]

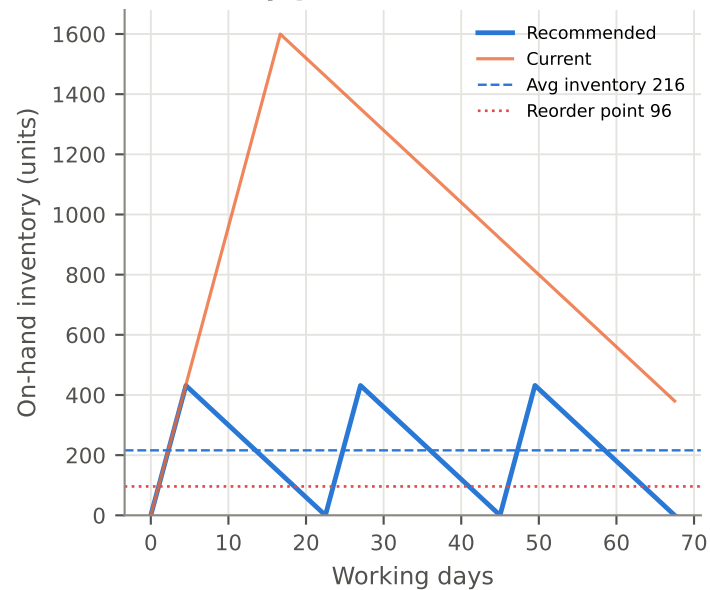
Total annual cost curve (solid = applicable price, dotted = extension)



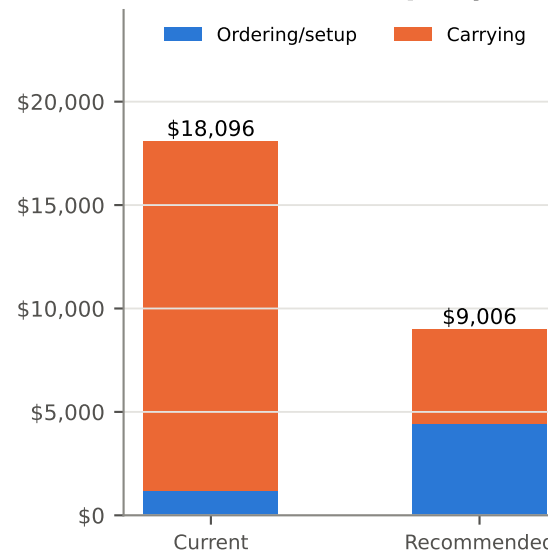
Key figures

Metric	Value
Annual demand D	6,000
Order/setup cost A	\$400.00
Base unit cost v0	\$96.000
Carrying rate r	0.22
Production rate m	30,000
Textbook EOQ/EPQ	533.0
Recommended Q	540
Price paid	\$96.000
Orders per year	11.1
Cycle (working days)	22.5
Reorder point	96
TRC (order+carry)	\$9,006
Total annual cost	\$585,006
Case / tier	-
Rules applied	Pack 20
Current Q	2,000
Annual savings	\$9,090
Current PCP	100.9%

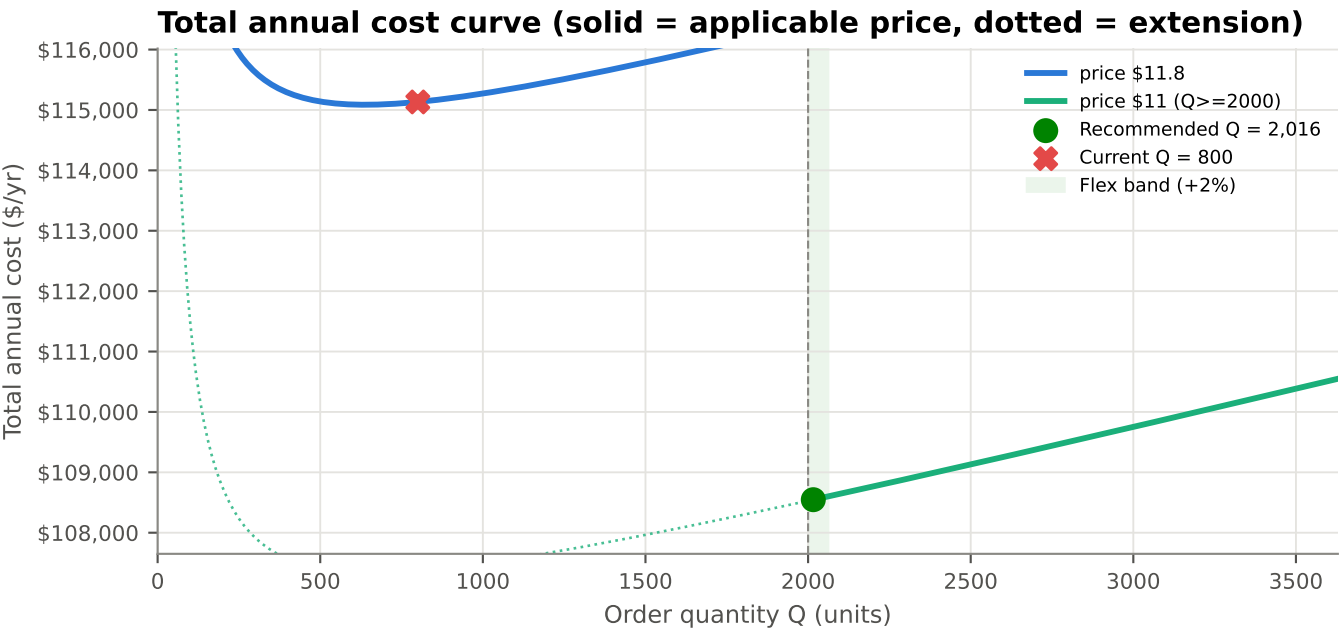
Inventory profile over time (sawtooth)



TRC breakdown (per year)

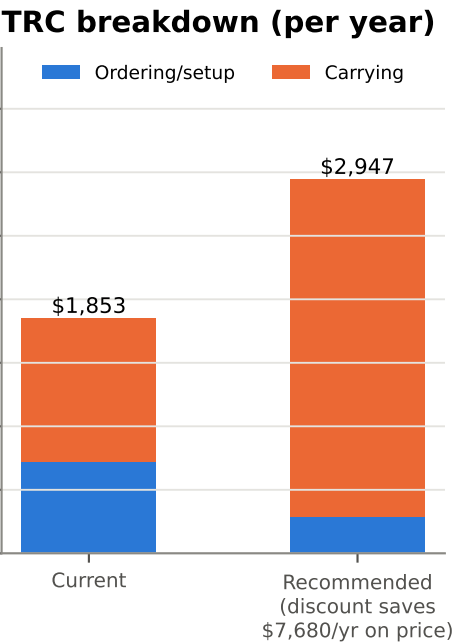
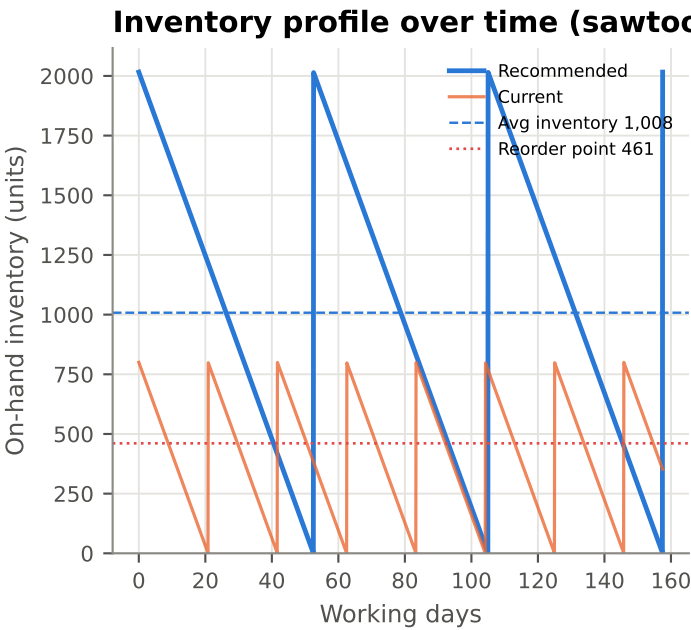


RM-1004 - Mechanical seal kit [EOQ + discounts]



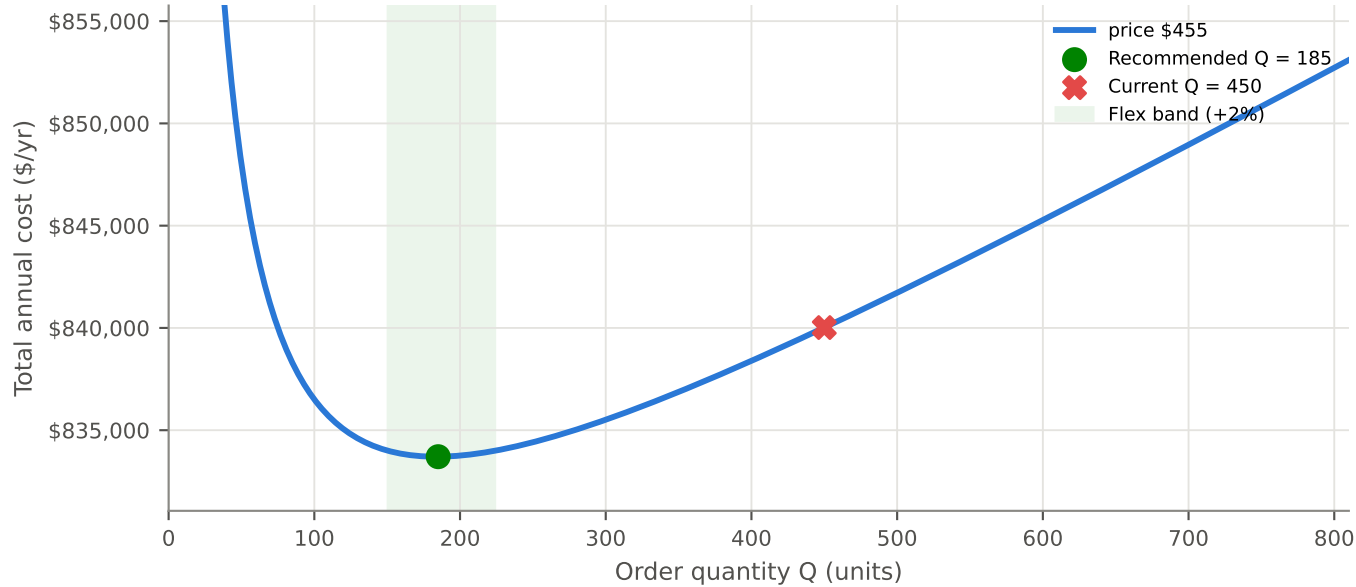
Key figures

Metric	Value
Annual demand D	9,600
Order/setup cost A	\$60.00
Base unit cost v_0	\$11.800
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	637.8
Recommended Q	2,016
Price paid	\$11.000
Orders per year	4.8
Cycle (working days)	52.5
Reorder point	461
TRC (order+carry)	\$2,947
Total annual cost	\$108,547
Case / tier	Case 1
Rules applied	Pack 24
Current Q	800
Annual savings	\$6,586
Current PCP	223.5%

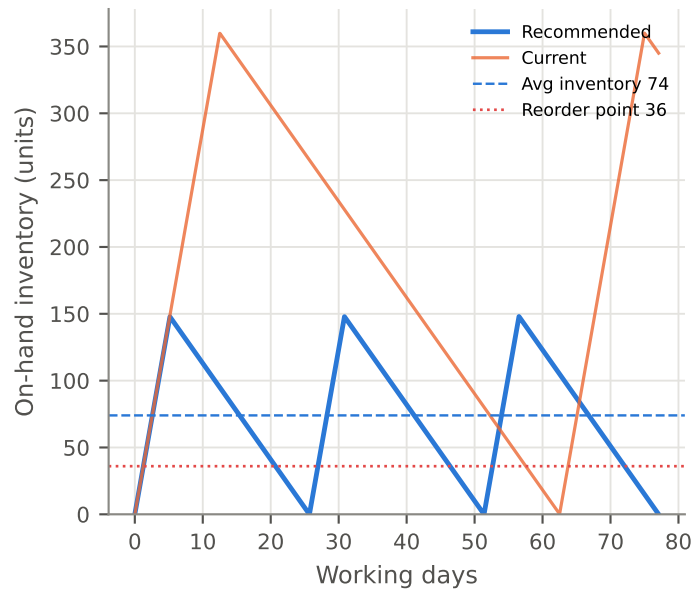


FG-3002 - Centrifugal pump CP-80 [EPQ]

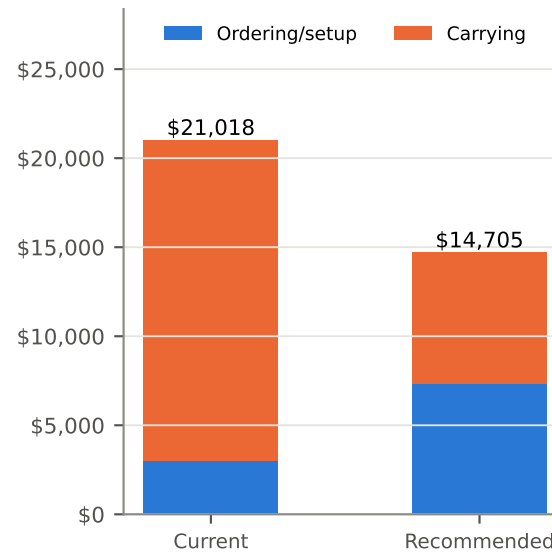
Total annual cost curve (solid = applicable price, dotted = extension)



Inventory profile over time (sawtooth)



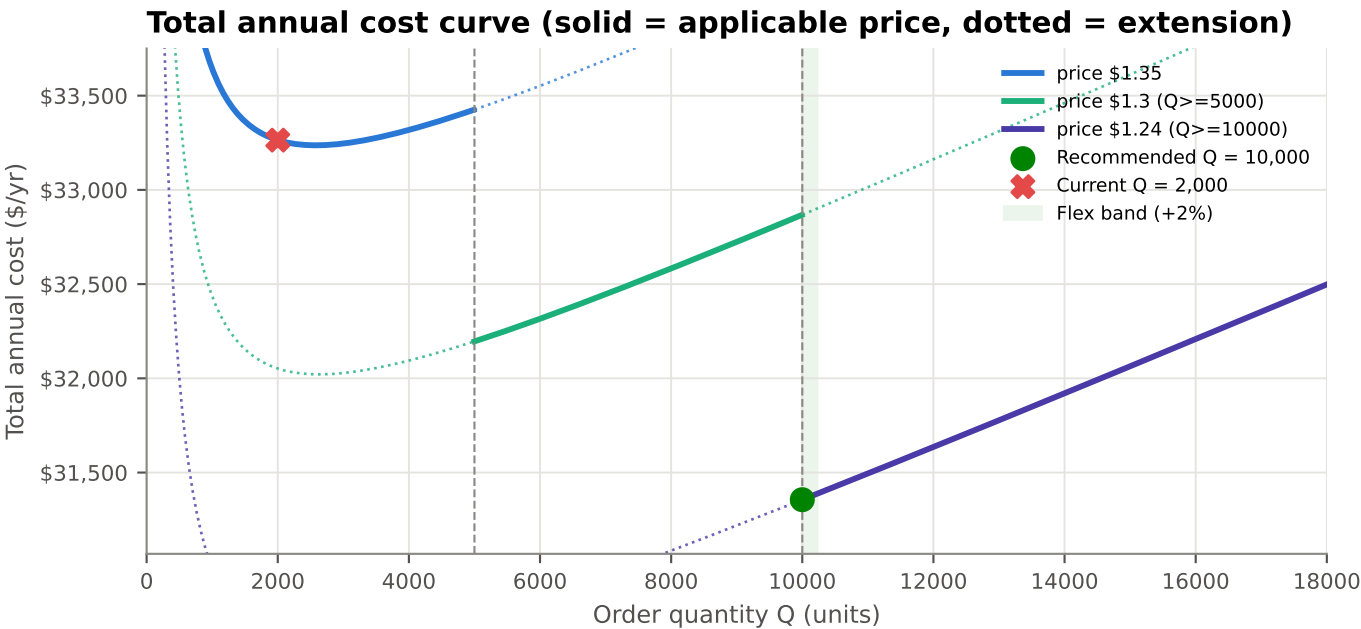
TRC breakdown (per year)



Key figures

Metric	Value
Annual demand D	1,800
Order/setup cost A	\$750.00
Base unit cost v0	\$455.000
Carrying rate r	0.22
Production rate m	9,000
Textbook EOQ/EPQ	183.6
Recommended Q	185
Price paid	\$455.000
Orders per year	9.7
Cycle (working days)	25.7
Reorder point	36
TRC (order+carry)	\$14,705
Total annual cost	\$833,705
Case / tier	-
Rules applied	Pack 5
Current Q	450
Annual savings	\$6,313
Current PCP	42.9%

RM-1005 - Ball bearing 6204-2RS [EOQ + discounts]



Key figures

Metric	Value
Annual demand D	24,000
Order/setup cost A	\$45.00
Base unit cost v0	\$1.350
Carrying rate r	0.24
Production rate m	-
Textbook EOQ/EPQ	2,582.0
Recommended Q	10,000
Price paid	\$1.240
Orders per year	2.4
Cycle (working days)	104.2
Reorder point	672
TRC (order+carry)	\$1,596
Total annual cost	\$31,356
Case / tier	Tier 3 of 3
Rules applied	Pack 50
Current Q	2,000
Annual savings	\$1,908
Current PCP	119.5%

